

SAUMYA SHAH

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EDUCATION

Quantum Informatics CDT

Doctor of Philosophy (PhD) in Quantum Informatics

United Kingdom

Sep 2026 – Aug 2030

- Joint programme; commencing at the University of Edinburgh.

National University of Singapore (NUS)

Bachelor of Computing in Computer Science (Honours)

Singapore

Aug 2022 – May 2026

- Specialisation in Artificial Intelligence | Minor in Astronomy
- **GPA:** 4.59/5.0 (Honours – Highest Distinction)
- **Awards & Scholarships:** Dean's List (School of Computing), NUS College President's Honour Roll, Certificate of Distinction in the AI Focus Area, NUS College International Scholar
- **Honours Thesis:** Identifying and Exploiting Physical Symmetries in Variational Quantum Many-Body Simulation.

PREPRINTS & PUBLICATIONS

- **Shah, S.**, Rebentrost, P. (2025). "Efficient Identification of Permutation Symmetries in Many-Body Hamiltonians via Graph Theory" [arXiv:2511.23160](#).
- **Shah, S.**, Khoo, Z.Y., Yang, A., Bressan, S. (2025). "Rediscovering the Lunar Equation of the Centre with AI Feynman via Embedded Physical Biases." Undergraduate Research Project, National University of Singapore. (Presented at the NUS UROP-Research Experience Conference 2025, Singapore) [arXiv:2511.09979](#).

RESEARCH EXPERIENCE

Symmetry-Aware Neural Network Quantum State Synthesis

Centre for Quantum Technologies, Singapore

Supervisor: Professor Patrick Rebentrost

Jun 2025 – May 2026

- Developed and proved novel graph-theoretic algorithms to find the permutation symmetry group in the Pauli and second quantisation bases with tractable (polynomial time) complexity.
- Introduced a novel polynomial-time Hamiltonian term partitioning algorithm with a mean 7% reduction in commuting groups compared to existing heuristics.
- Introduced the *Group-Equivariant Transformer Quantum State* architecture to efficiently find the ground state of symmetric systems.
- Showed that the GETQS demonstrates parameter efficiency and accelerated convergence compared to the baseline.

Multi-Agent RL for Football Strategy Optimisation

NUS School of Computing, Singapore

Supervisor: Dr Amirhassan Monajemi

Sep 2025 – Nov 2025

- Applied reinforcement learning and generative AI techniques to model and optimise tactical decisions in professional football, training agents on match data and simulated environments.
- Evaluated strategies, predicted outcomes of tactical choices, and generated adaptive simulations in the Google Research Football environment to support coaches in refining team performance.
- Created and presented a poster at NUS Artificial Intelligence Institute's AI+X Conference.

Physics-Informed Symbolic Regression

NUS School of Computing, Singapore

Supervisors: Professor Stéphane Bressan[†], Dr Abel Yang

Dec 2023 – Jan 2025

- Investigated physics-informed machine learning algorithms for symbolic regression on astronomical datasets, such as AI Feynman.
- Introduced physics-informed inductive and observational biases to AI Feynman to successfully recover the Equation of the Centre from Lunar Ephemerides data.
- Developed an extension to automatically find a canonical coordinate system transformation for AI Feynman to generate a set of solutions with improved parsimony and accuracy.
- Findings were orally presented at the NUS UROP-Research Experience Conference 2025, and summarised in a manuscript available on ArXiv.

INDUSTRY & TEACHING EXPERIENCE

Technical Assistant, <i>HS1503 (Understanding AI)</i>	NUS, Singapore, <i>Oct 2025 – May 2026</i>
Teaching Assistant, <i>CS2109S (AI & Machine Learning)</i>	NUS, Singapore, <i>Aug 2024 – Dec 2024</i>
Student Assistant, <i>GEA1000N (Quantitative Reasoning)</i>	NUS, Singapore, <i>Aug 2023 – Aug 2024</i>

SELECTED PROJECTS

- **QuantIron** ([Github Link](#)) | A high-performance, hardware-accelerated modular quantum computing library with a focus on physical applications built in Rust.
- **DeepIron** ([Github Link](#)) | A machine learning library with deep learning features built from scratch in Rust.
- **LiveIron** ([Github Link](#)) | A performant, extensible cellular automata simulator library with visualisation capabilities built in Rust. Supports evolving cellular automata using genetic algorithms (Mitchell et al, 1996).
- **RustyKAN** ([Github Link](#)) | A library allowing for easy and performant implementation of Kolmogorov-Arnold Networks, built from scratch in Rust (Liu et al, 2024).

SERVICE

General Secretary, <i>Staff Advisor: Dr Abel Yang</i>	NUS Astronomical Society, Singapore, <i>Aug 2023 – Aug 2026</i>
Volunteer, <i>Supervisor: Dr Shaun Lin</i>	Impact Experience Project, Cambodia, <i>Aug 2023 – May 2025</i>